

**Generic Scenario: Electroplating**  
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Prepared for the Chemical Engineering Branch by SAIC

## **Introduction**

Electroplating is the process by which a metal coating is applied to an object immersed in an electrolytic solution by passing a current through the solution. The article to be plated is the cathode. The plating metal is the anode, and when power is applied, it is oxidized and dissolved in the solution. The ions in solution are then reduced, or plated, on the cathode. When the surface to be plated is the anode, the process is called anodizing. Electroless or autocatalytic plating, uses a chemical reducing agent to reduce a dissolved metal onto a surface without the use of an electric current. This scenario will handle these similar processes.

Any electrically conductive surface can be electroplated. The most common metals plated on these surfaces include cadmium, chromium, cobalt, copper, gold, iron, lead, nickel, silver, tin, zinc and many alloys. The various uses for electroplating include: corrosion resistance, decoration, worn parts salvage, paint base, barrier coating, and electronic circuitry.

The process begins with several pre-treatment cleaning steps including immersion and spray rinses with deionized water and alkaline or acidic solutions. The parts are then dipped into the plating bath and then rinsed by several more immersion and spray rinses with deionized and permeate water. Lastly the part is baked to cure the coating. Although electroplating includes all of the above-mentioned stages, this generic scenario is applicable only to plating followed by rinsing.

## **Market**

The number of electroplating shops dropped from 12,000 to 7,500 between the years of 1980 and 1992 due to the difficulty of meeting waste regulations imposed upon stream effluents (KO, 1994). This reduction impacted the smaller job shops moreso than the larger electroplating shops. One indicator for gauging the market is anode sales.

Electroplating in the automobile industry is declining due to the replacement of some metal parts by plastics, but the plumbing and appliance sectors are allowing for moderate growth with the increased use of electroless and other functional plating. The printed circuit board industry adds another market for the use of electroplating, as well as off-shore drilling in Texas. (USEPA, 1984)

## **Use-Industrial Electroplating**

Electroplating is an estimated 4.5 billion dollar industry (BOC, 1995) with applications ranging from automobiles and sporting goods to electronics and jewelry. Approximately 14.2 million ft<sup>2</sup> of metal are plated each year (USEPA, May 1995). The industry produces 557,000 lb/yr of priority metal pollutants along with 1.195 billion gallons of wastewater generated from the unit operations and rinse water of the process (USEPA, May 1995). Additionally, chromium was chosen in determining an average bath size based on its prevalence in industry and on its wide range of uses in both decorative and functional plating. Bath size varies significantly however,

according to several parameters including the type of bath and the size and shape of the immersed objects. The size of a bath can range from just a liter to over 100,000 liters. The automobile industry uses tanks of 250,000 liters in size (KO, 1994).

### Process Description

Parts are dipped into a plating bath, which consists of the metal ions in solution (as provided by dissolved metal from the anode or metal salts/oxides), complexing agents, stabilizers to prevent hydrolysis, buffers for pH maintenance, and catalysts (USEPA, May 1995). These chemicals range from amounts measured in ppm for brighteners or other additives to 20% for metal salts. The additives are nonvolatile and tend to remain in the bath. The baths, however, are equipped with exhaust systems for any vapors that may exist above the solution. Electroplating baths are almost always aqueous-based. The dip time and bath size depend on the metal to be plated, the thickness of the desired coating, and the size and shape of the immersed object. After dipping is complete, the rinsing stages begin.

For Plating baths:

Number of use sites:

$$NS = PV \text{ (kg/yr)} / [((\text{gallons of bath solution/site-yr}) \times (3.78 \text{ l/gal}) \times (\text{Concentration of PMN in bath})) + ((\text{gallons of rinsewater produced/site-yr}) \times (3.78 \text{ l/gal}) \times (\text{concentration of PMN in rinsewater}) \times (1 - \text{waste water recovery efficiency}))]$$

Assume 15,570 gallons of plating bath solution per site-yr (basis: 14.9 million gallons of plating bath water are generated from 957 sites per year (U.S.EPA, May, 1995))

Assume 1.233 million gallons of rinse water produced per site-yr (basis: U.S.EPA, May, 1995)

Assume concentration of PMN in rinsewater is half of that in bath (basis: the metals concentration in rinsewaters is found as high as 23,500 mg/l and as high as 43,400 mg/l in the bath itself, or about half the concentration (USEPA, May 1995))

Assume a recovery efficiency of 90% (basis: a valid recovery of plating bath solution from rinse water can recover 90% of the value lost (Durney, 1984))

Days/year Operation

Assume 250

(basis: 50,300 people perform production work in electroplating and polishing for a total of 103,900,000 hours (BOC, 1995). At 8 hr/d, there are 258 days of operation.

## Environmental Releases

### Water

The electroplating industry discharges 1,195 million gallons of water per year, the majority of which, 1,180 million gallons per year, is used for the electroplating rinses (USEPA, May 1995). With 957 sites performing electroplating unit operations (USEPA, May 1995), there is, on average, 4932 gal/site-day of rinse water and 62 gal/site-day of electroplating wastewaters discharged. The bath is continuously purified at about 1 bath turnover per hour to remove metal salt impurities and organic impurities (Durney, 1984). The metal ions and additives necessary for plating are returned to the bath with the filtered bath water solution. The additives are therefore discharged with rinsewaters or spent bath disposals in the form of wastewater or sludge.

#### *Daily water releases due to unit operations and rinses*

Releases, kg/site-day = Releases from rinses/day over 250 days/yr

Daily releases from rinses, kg/site-day = (Volume of rinsewater generated/site-day) x (Conc. of PMN in rinsewater) x (1 - recovery efficiency) x (1 - wastewater treatment efficiency)

Assume 18,600 liters (4932 gallons) of wastewater generated/site-day  
(basis: USEPA, May 1995)

Assume a recovery efficiency of 90% (basis: a valid recovery of plating bath solution from rinse water can recover 90% of the value lost (Durney, 1984))

Assume concentration of PMN in rinsewater is half of that in bath (basis: the metals concentration in rinsewaters is found as high as 23,500 mg/l and as high as 43,400 mg/l in the bath itself, or about half the concentration (USEPA, May 1995))

Assume a raw waste water treatment efficiency of 97% for suspended solids and 50% for organic compounds (basis: The average raw waste concentration of 78 sites for total suspended solids was 600 mg/L and that for the effluent was 16 mg/L resulting in a removal of 97%. The average removal efficiency for total organic content for five samples was 45.2% using reverse osmosis (U.S.EPA, 5/1995))

#### *Biannual water releases due to disposal of spent bath*

Releases, kg/site-yr = Releases from spent bath over 2 days/yr.

Yearly release, kg/site-yr = (Bath volume) x (Dumping frequency) x (PMN bath concentration) x (1 - waste water treatment efficiency)

Assume the bath volume of one site is 7500 gallons total (basis: the size and number of tanks per chromium plating facility were averaged for 17 of 66 documented sites yielding: 2500 gal/tank with 3 tanks/plant (USEPA, April 1995))

Assume Dumping frequency of six months (basis: 14.9 million gallons of wastewater from electroplating baths are generated/yr. With 957 electroplating sites and 7500 gal of bath material/site, the entire bath would be dumped twice/yr (U.S.EPA, May 1995))

Assume a raw waste water treatment efficiency of 97% for suspended solids and 50% for organic compounds (basis: The average raw waste concentration of 78 sites for total suspended solids was 600 mg/L and that for the effluent was 16 mg/L resulting in a removal of 97%. The average removal efficiency for total organic content for five samples was 45.2% using reverse osmosis (U.S.EPA, 5/1995))

#### Air

For vapor pressure < 0.001 torr, air releases of the baths are assumed to be negligible. (Basis: The estimated high end for releases to air of an additive at one percent concentration due to mist from chromium baths was less  $10^{-6}$  kg/day)

For vapor pressure > 0.001 torr, see CEB models for open tank.

#### Landfill/Incineration

Yearly discharge to sludge = Sludge losses due to disposal of spent bath + Sludge losses due to rinsewater disposal:

Yearly release, kg/site-yr = (waste water treatment filter efficiency) [(Volume of bath) x (Dumping frequency) x (PMN bath concentration) + (Volume of rinse water) x (1 - rinse water recovery efficiency) x (Conc. of PMN in rinsewater)]

Assume a waste water treatment efficiency of 97% for suspended solids and 50% for organic compounds (basis: The average raw waste concentration of 78 sites for total suspended solids was 600 mg/L and that for the effluent was 16 mg/L or a removal of 97%. The average removal efficiency for total organic content for five samples was 45.2% using reverse osmosis (U.S.EPA, 5/1995))

Assume 1.233 million gallons of wastewater generated/site-yr (basis: USEPA, May 1995)

Assume a recovery efficiency of 90% (basis: a valid recovery of plating bath solution from rinse water can recover 90% of the value lost (Durney, 1984))

Assume concentration of PMN in rinsewater is half of that in bath (basis: the metals concentration in rinsewaters is found as high as 23,500 mg/l and as high as 43,400 mg/l in the bath itself, or about half the concentration (USEPA, May 1995))

### Occupational Exposure

Electroplating personnel may be classified as line operators, lead persons, maintenance, or laboratory technicians. The amount of exposure depends on the job description. Operators and lead persons walk the line, operate the controls, load and unload plated objects, and repair minor malfunctions. The exposure time for maintenance crews varies with the frequency of system malfunctions. Laboratory technicians perform one quality control assessment per day and are responsible for chemically recharging the tanks once or twice per week. (HETA 81-439-1256)

Days/year Operation:

Assume 250

(50,300 production workers work in electroplating and polishing for a total of 103,900,000 hours (BOC, 1995). At 8 hr/d, there are 258 days of operation.)

Hours/site-day:

Assume 8

Number of workers/site

20 workers/site

Assume number of workers/site for electroplating and polishing sites is similar to number of workers/site at electroplating sites alone

(basis: 65,400 total employees in electroplating and polishing at 3,296 total establishments yielding 19.8 employees/site (BOC, 1995))

### Inhalation Exposure:

Inhalation exposure is a result of evolved hydrogen and oxygen gas rising from the submerged part, the anode, or the cathode, and dragging bath solution with it into the air.

For PMN exposure during electroplating,

Potential dose rate (mg/d),  $I = C_m b h$

Where:

$I$  = Daily inhalation exposure, mg/day

$C_m$  = Airborne concentration of PMN, mg/m<sup>3</sup>

$b$  = Inhalation rate, m<sup>3</sup>/hr

$h$  = Duration, hr/day

Assume inhalation rate of 1.25 m<sup>3</sup>/hr (basis: typical worker breathing volume (CEB, 1991))

Assume 8 hr duration (BOC, 1995)

To obtain  $C_m$

Assume for mist, the airborne concentration of PMN can be expressed as:

$$C_{m,PMN} = C_{m,k} Y_{PMN}/Y_k \text{ (basis: CEB, 1991)}$$

Where:  $C_{m,PMN}$  = Estimated airborne concentration of the PMN chemical,  $\text{mg}/\text{m}^3$   
 $C_{m,k}$  = Measured airborne concentration of the known chemical  
 $Y_{PMN}$  = Weight fraction of PMN chemical in mixture  
 $Y_k$  = Weight fraction of known chemical in mixture

Assume airborne concentration to be the maximum allowed OSHA limit (see Chart B for airborne limits for constituents typically found in metal plating baths). For default value, choose the OSHA value for chromium of  $0.5 \text{ mg}/\text{m}^3$ .

Typical values for metal concentrations in electroplating baths can also be found in Chart B. For a default value, use the chromium bath weight fraction of 0.25.

For volatile PMN's (vapor pressure > 0.001 torr) use the CEB models for an open tank.

#### Dermal exposure:

To estimate dermal exposure to the PMN during sampling of baths or the measurement and addition of bath replenishment chemicals, the dermal contact model presented in the CEB manual should be used with the following assumptions for incidental 2-hand contact (CEB, 1991):

$$D = SQC$$

Where:  $D$  = Dermal Exposure ( $\text{mg}/\text{day}$ )  
 $S$  = Surface area of contact ( $\text{cm}^2$ ) = 1300  
 $Q$  = Quantity typically remaining on the skin ( $\text{mg}/\text{cm}^2$ ) = 1-3  
 $C$  = PMN concentration in bath

#### Uncertainties/Data Gaps

Data for releases tend to be available for metals only. Most PMN's however, are additives and therefore the filter efficiencies and volatilities will vary. Filter efficiencies were determined for broad classes such as "total suspended solids" or "total toxic organics" and it was assumed that relative concentrations of constituents in all streams and in the mist would be constant.

When the metal salt impurities and organic impurities are removed from the bath by the continuous purification filter, some types of organic wetting and brightening agents may be removed to some degree. Without information on a given PMN, it is impossible to predict the releases of PMN from the continuous filtration system without more information on its solubility

in water. For this reason, it is assumed that releases of PMN occur only from spent bath disposal and from rinsewater.

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Chart B, Metal Concentrations

| Metal  | Bath type          | Bath Conc.s <sup>1</sup> , g/L                                                                                                                                    | Conc. above bath, mg/m <sup>3</sup> |
|--------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Brass  | Yellow, regular    | CuCN: 32<br>Zn(CN) <sub>2</sub> : 10<br>NaCN: 50<br>Na <sub>2</sub> CO <sub>3</sub> : 7.5<br>NaHCO <sub>3</sub> : 10<br>NaOH: --<br>NH <sub>4</sub> OH: 2.5-5mL/L |                                     |
|        | Yellow, high speed | CuCN: 75<br>Zn(CN) <sub>2</sub> : 5<br>NaCN: 125<br>Na <sub>2</sub> CO <sub>3</sub> : --<br>NaHCO <sub>3</sub> : --<br>NaOH: 45<br>NH <sub>4</sub> OH:            |                                     |
|        | White              | CuCN: 10<br>Zn(CN) <sub>2</sub> : 60<br>NaCN: 100<br>Na <sub>2</sub> CO <sub>3</sub> : 40<br>NaHCO <sub>3</sub> : --<br>NaOH: 38<br>NH <sub>4</sub> OH: -         |                                     |
| Bronze | Typical            | CuCN: 29-35<br>Cu metal: 20-25<br>Na <sub>2</sub> SnO <sub>3</sub> .3H <sub>2</sub> O: 35-38<br>Sn metal: 14-17<br>NaCN: 54-64<br>NaOH: 7.5-10                    |                                     |
|        | Speculum           | CuCN: 11<br>Cu metal: 8<br>Na <sub>2</sub> SnO <sub>3</sub> .3H <sub>2</sub> O: 90 Sn metal: 40<br>NaCN: 27<br>NaOH: 16                                           |                                     |

|             |               |                                                                                                                                                                                      |                                          |
|-------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Cd          | Cyanide bath  | Cd <del>metal</del> : 20-30<br>NaCN: 90-150<br>Na <sub>2</sub> CO <sub>3</sub> : 30-60<br>H <sub>2</sub> SO <sub>4</sub> :—<br>NH <sub>4</sub> BF <sub>4</sub> :—<br>NaOH:10-20      | NIOSH: 0.04<br>(NIOSH, 1984)             |
|             | Acid sulfate  | Cd <del>metal</del> : 15-30<br>NaCN: —<br>Na <sub>2</sub> CO <sub>3</sub> : —<br>H <sub>2</sub> SO <sub>4</sub> :45-90<br>NH <sub>4</sub> BF <sub>4</sub> :—<br>NaOH:—               |                                          |
|             | Fluoro-borate | Cd <del>metal</del> :75-150<br>NaCN: —<br>Na <sub>2</sub> CO <sub>3</sub> : —<br>H <sub>2</sub> SO <sub>4</sub> :—<br>NH <sub>4</sub> BF <sub>4</sub> :60-120<br>NaOH:—              |                                          |
| Cr, typical | Conventional  | CrO <sub>3</sub> : 240-260<br>SO <sub>4</sub> <sup>2-</sup> : 2.4-2.6<br>SiF <sub>6</sub> <sup>2-</sup> : —<br>CrO <sub>3</sub> :SO <sub>4</sub> <sup>2-</sup> ratio: 90-110:1       | OSHA: 0.5<br>HETA-87-353-1899<br>5/88    |
|             |               | CrO <sub>3</sub> : 150-180<br>SO <sub>4</sub> <sup>2-</sup> : 0.9-1.0<br>SiF <sub>6</sub> <sup>2-</sup> : 0.5-0.6<br>CrO <sub>3</sub> :SO <sub>4</sub> <sup>2-</sup> ratio:170-180:1 |                                          |
| Cu, cyanide | CN, strike    | Cu <del>metal</del> : 15-22<br>CuCN: 21-31<br>KCN: 31-70<br>KOH: 3-18<br>Rochelle salts: 10-20<br>Na <sub>2</sub> CO <sub>3</sub> : 10-15                                            | OSHA: 0.5<br>(HETA-87-353-1899,<br>5/88) |
|             | Rochelle      | Cu <del>metal</del> : 22-36<br>CuCN: 31-51<br>KCN: 55-89<br>KOH: 12-18<br>Rochelle salts: 15-25<br>Na <sub>2</sub> CO <sub>3</sub> : 30-45                                           |                                          |

|                    |                        |                                                                                                                                                                                                                                                                                               |  |
|--------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                    | High speed             | Cu metal: 56-71<br>CuCN: 79-100<br>KCN: 130-165<br>KOH: 20-25<br>Rochelle salts: 15-25<br>Na <sub>2</sub> CO <sub>3</sub> : 35-55                                                                                                                                                             |  |
| Cu, acid           | Sulfate, average range | Cu metal: 38-64<br>CuSO <sub>4</sub> .5H <sub>2</sub> O: 150-250<br>H <sub>2</sub> SO <sub>4</sub> : 30-75<br>Cl <sup>-</sup> : 0.20-0.12<br>Cu(BF <sub>4</sub> ) <sub>2</sub> : --<br>HBF <sub>4</sub> : --<br>H <sub>3</sub> BO <sub>3</sub> : --                                           |  |
|                    | Fluoroborate, range    | Cu metal: 60-120<br>CuSO <sub>4</sub> .5H <sub>2</sub> O: -- H <sub>2</sub> SO <sub>4</sub> :<br>--<br>Cl <sup>-</sup> : --<br>Cu(BF <sub>4</sub> ) <sub>2</sub> : 225-450<br>HBF <sub>4</sub> : 15-30<br>H <sub>3</sub> BO <sub>3</sub> : 15-30                                              |  |
| Cu, Pyro-phosphate | Typical                | Cu metal: 19-30<br>Cu <sub>2</sub> P <sub>2</sub> O <sub>7</sub> .3H <sub>2</sub> O: 53-84<br>K <sub>4</sub> P <sub>2</sub> O <sub>7</sub> .3H <sub>2</sub> O: 235-405<br>NH <sub>4</sub> OH: 3.75-11<br>KNO <sub>3</sub> : 3.0-6.0<br>P <sub>2</sub> O <sub>5</sub> : Cu ratio:<br>7.0-7.5:1 |  |
| Au.                | Alkaline               | Au metal: 2-12<br>KAu(CN) <sub>2</sub> : 3-18<br>KCN: 15-48<br>K <sub>2</sub> CO <sub>3</sub> : 0-45<br>K <sub>2</sub> HPO <sub>4</sub> : 0-45<br>KOH: 1-30                                                                                                                                   |  |
|                    | Neutral                | Au metal: 4-16<br>KAu(CN) <sub>2</sub> : 6-24<br>KCN: --<br>K <sub>2</sub> CO <sub>3</sub> : --<br>K <sub>2</sub> HPO <sub>4</sub> : 0-90<br>KOH: --                                                                                                                                          |  |

|    |                     |                                                                                                                                                                                                                                 |                                                          |
|----|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
|    | Acid                | Au metal: 2-16<br>$\text{KAu}(\text{CN})_2$ : 3-24<br>KCN: -<br>$\text{K}_2\text{CO}_3$ : -<br>$\text{K}_2\text{HPO}_4$ : 0-100<br>KOH: -                                                                                       |                                                          |
|    | Strike              | Au metal: 0.5-2<br>$\text{KAu}(\text{CN})_2$ : 0.75-3<br>KCN: 15-90<br>$\text{K}_2\text{CO}_3$ : -<br>$\text{K}_2\text{HPO}_4$ : 15-45<br>KOH: -                                                                                |                                                          |
| Ni | Watts               | Ni metal: 82<br>$\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$ : 300<br>$\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ : 60<br>$\text{Ni}(\text{SO}_3\text{NH}_2)_2 \cdot 4\text{H}_2\text{O}$ :<br>-<br>$\text{H}_3\text{BO}_3$ : 35-45  | OSHA: 1.0<br>NIOSH: 0.015<br>(HETA-81-439-1256;<br>1/83) |
|    | Watts high chloride | Ni metal: 77<br>$\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$ : 135<br>$\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ : 190<br>$\text{Ni}(\text{SO}_3\text{NH}_2)_2 \cdot 4\text{H}_2\text{O}$ :<br>-<br>$\text{H}_3\text{BO}_3$ : 35-45 |                                                          |
|    | Sulfamate           | Ni metal: 75<br>$\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$ : --<br>$\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ : -<br>$\text{Ni}(\text{SO}_3\text{NH}_2)_2 \cdot 4\text{H}_2\text{O}$ :<br>410<br>$\text{H}_3\text{BO}_3$ : 35-45  |                                                          |
| Ag | Decorative          | AgCN: 30-35<br>KCN: 50-78<br>$\text{K}_2\text{CO}_3$ : 15<br>$\text{KNO}_3$ : -<br>KOH: -                                                                                                                                       | OSHA: 0.01<br>(HHE-78-53-510<br>7/78)                    |
|    | Industrial          | AgCN: 45-50<br>KCN: 65-72<br>$\text{K}_2\text{CO}_3$ : 45-50<br>$\text{KNO}_3$ : 40-80<br>KOH: 40-60                                                                                                                            |                                                          |

|       |               |                                                                                                        |                               |
|-------|---------------|--------------------------------------------------------------------------------------------------------|-------------------------------|
|       | Strike        | AgCN: 1.5-5.0<br>KCN: 75-90<br>K <sub>2</sub> CO <sub>3</sub> : --<br>KNO <sub>3</sub> : --<br>KOH: -- |                               |
| Zn-Co | Acidic        | Zn: 25<br>Co: 4<br>Ni: --<br>Fe: --<br>chloride: 135<br>boric acid: 25<br>NaOH: --                     |                               |
|       | Alkaline      | Zn: 8<br>Co: 0.04<br>Ni: --<br>Fe: --<br>chloride: --<br>boric acid: --<br>NaOH: 90                    |                               |
| Zn-Ni | Acidic        | Zn: 30<br>Co: --<br>Ni: 25<br>Fe: --<br>chloride: 240<br>boric acid: --<br>NaOH: --                    |                               |
|       | Alkaline      | Zn: 8<br>Co: --<br>Ni: 1.6<br>Fe: --<br>chloride: --<br>boric acid: --<br>NaOH: 130                    |                               |
| Zn-Fe | Alkaline only | Zn: 8<br>Co: --<br>Ni: --<br>Fe: 0.05<br>chloride: --<br>boric acid: --<br>NaOH: 90                    | T.L.V.: 1.0<br>(Durney, 1984) |

1 Unless otherwise noted, bath concentrations were taken from KO, 1994

Chemical Exposure in Metalplating:

| Industry                  | Chemical          | Activity                | Exposure                      | OSHA standard                        | Reference              |
|---------------------------|-------------------|-------------------------|-------------------------------|--------------------------------------|------------------------|
| Industry Steel & Wire Co. | Chemical Chromate | Activity Normal Plating | Exp 0.035mg/m <sup>3</sup>    | OSHA standard                        | HHIE 73-111-125, 3/74  |
| Electro-metals            | Chromium          | Electro-plating         | 0.04-0.20 mg/m <sup>3</sup>   | 0.5 mg/m <sup>3</sup>                | HHE 80-156-747, 10/80  |
| Sporting goods            | Chromium          | Anodizing               | 0.01 mg/m <sup>3</sup>        | 0.5 mg/m <sup>3</sup>                | HHIE 79-23-603, 7/79   |
| Chrome Plating            | Chromium VI       | Electro-plating         | 0.003-0.006 mg/m <sup>3</sup> | 0.1 mg/m <sup>3</sup>                | HHE 81-085-889, 6/81   |
| Chrome Plating            | Chromium (Total)  | Electro-plating         | 0.009-0.81 mg/m <sup>3</sup>  | 0.5 mg/m <sup>3</sup>                | HHE 81-085-889, 6/81   |
| Steel & Wire Co.          | Cyanide           | Normal plating          | 0.08 mg/m <sup>3</sup>        |                                      | HHIE 73-111-125, 3/74  |
| Auto-motive               | Fluoride          | Plating Dep't           | 0.072 mg/m <sup>3</sup>       |                                      | HETA-81-439-1256, 1/81 |
| Steel & Wire Co.          | HCl               | Normal plating          | 0.007 mg/m <sup>3</sup>       |                                      | HHE 73-111-125, 3/74   |
| Sporting goods            | Naptha            | Painting& Cleaning      | 7-100 mg/m <sup>3</sup>       | 400 mg/m <sup>3</sup>                | HHIE 79-23-603, 7/79   |
| Auto-motive               | Nickel            | Electro-plating         | 0.005-0.008 mg/m <sup>3</sup> | 1.00 mg/m <sup>3</sup> (NIOSH-0 015) | HETA 81-439-1256, 1/83 |

|                 |               |                           |                                                                   |                                                                |                        |
|-----------------|---------------|---------------------------|-------------------------------------------------------------------|----------------------------------------------------------------|------------------------|
| Electro-plating | Nickel        | Electro-plating           | Acute exposure: 0.5 to 2.5 g                                      |                                                                | Sundermanet. al., 1988 |
| Electro-plating | Nickel empds  | Electro-plating           | Urine: 0.7-50 mu g/l<br>hands: 1.9-547 mu g<br>face: 1.0 -86 mu g |                                                                | Bavazzanoet. al., 1994 |
| Electro-plating | Silver        | Weighing & Preparing bath | 0.04 mg/m <sup>3</sup>                                            | <0.002 (2 room samples) & 0.01 mg/m <sup>3</sup> close to bath | HHE-78-53-510, 7/78    |
| Chrome plating  | Sulfuric Acid | Electro-plating           | 0.010-0.015 mg/m <sup>3</sup>                                     | 1 mg/m <sup>3</sup>                                            | IIIE 81-085-889, 6/81  |